McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 8
Soil Moisture Sensing and Ubidots IoT Platform

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B. Tech - SMRTTECH

OBJECTIVES

In this lab, you will explore some new features of Ubidots IoT Platform. An **Arduino device** will be interfaced with **a capacitive soil moisture sensor** that is typically used in **IoT gardening** applications. Through LabVIEW, the sensor readings will be published to Ubidots MQTT broker. You will configure Ubidots to accomplish the following functions:

1. Display moisture readings.
2. Set Moisture Thresholds.
3. Send email alerts.
4. Display local weather (optional).

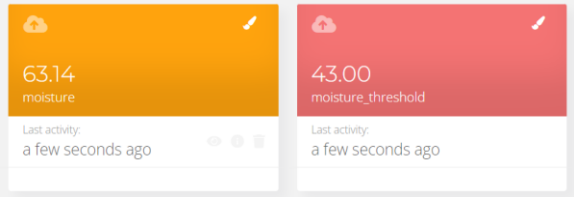
MATERIALS

- Arduino Mega
- Capacitive Soil Moisture Sensor
- Cup of water

PROCEDURE

Exercise 1: Ubidots IoT Platform Arduino Device Configuration

1. Login to your Ubidots account using: https://industrial.ubidots.com/accounts/signin/
2. Go to **Devices»Arduino**,
    a. delete all the variables of Lab 7
    b. create two new raw variables: **moisture** and **moisture_threshold**
    c. set the color of the moisture_threshold variable to red.

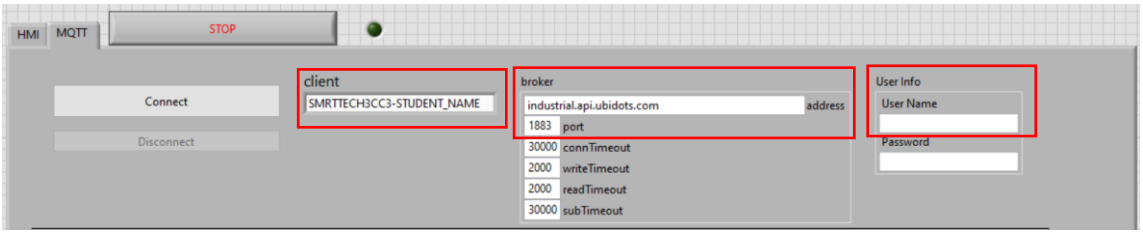


Hint: for more details, please refer to page 3 of lab 7 manual.

End of Exercise

Exercise 2: Exchanging Data between LabVIEW and Ubidots MQTT broker

1. From avenue to learn download Lab8.zip files. Extract the zip file. Open the extracted folder and open LabVIEW file named "Lab8_init.vi".
2. On the front panel, click the MQTT Tab and configure the client name, the broker address and port, and the user name as described in Page 4 of Lab 7 manual.



3. Modify the published topic to be **/v1.6/devices/arduino/moisture**.
4. Modify the subscibe topic to be **/v1.6/devices/arduino/moisture_threshold/lv**.
5. Run the LabVIEW program, press Connect, Publish, Enable Auto Publish, and Subscribe push buttons. Confirm that all LED indicators are green and that the ValueSent and the RX Topic Indicators are not empty.

IMPORTANT NOTES

1. **Please do not stop your program by pressing the abort button . You should always stop your program by pressing the "Stop" Push button.**
2. **IN THIS LAB, THE STOP BUTTON STOP IS MAINTAINED. EVERYTIME, YOU WILL STOP THE PROGRAM BY PRESSING THE BUTTON, YOU MUST PRESS IT AGAIN BEFORE YOU RUN THE PROGRAM AGAIN.**

End of Exercise

Exercise 3: Create a moisture sensor circuit

1. Have a quick look at the capacitive sensor data sheet.

https://media.digikey.com/pdf/Data%20Sheets/DFRobot%20PDFs/SEN0193_Web.pdf

2. Build the circuit shown in **Figure 1**.

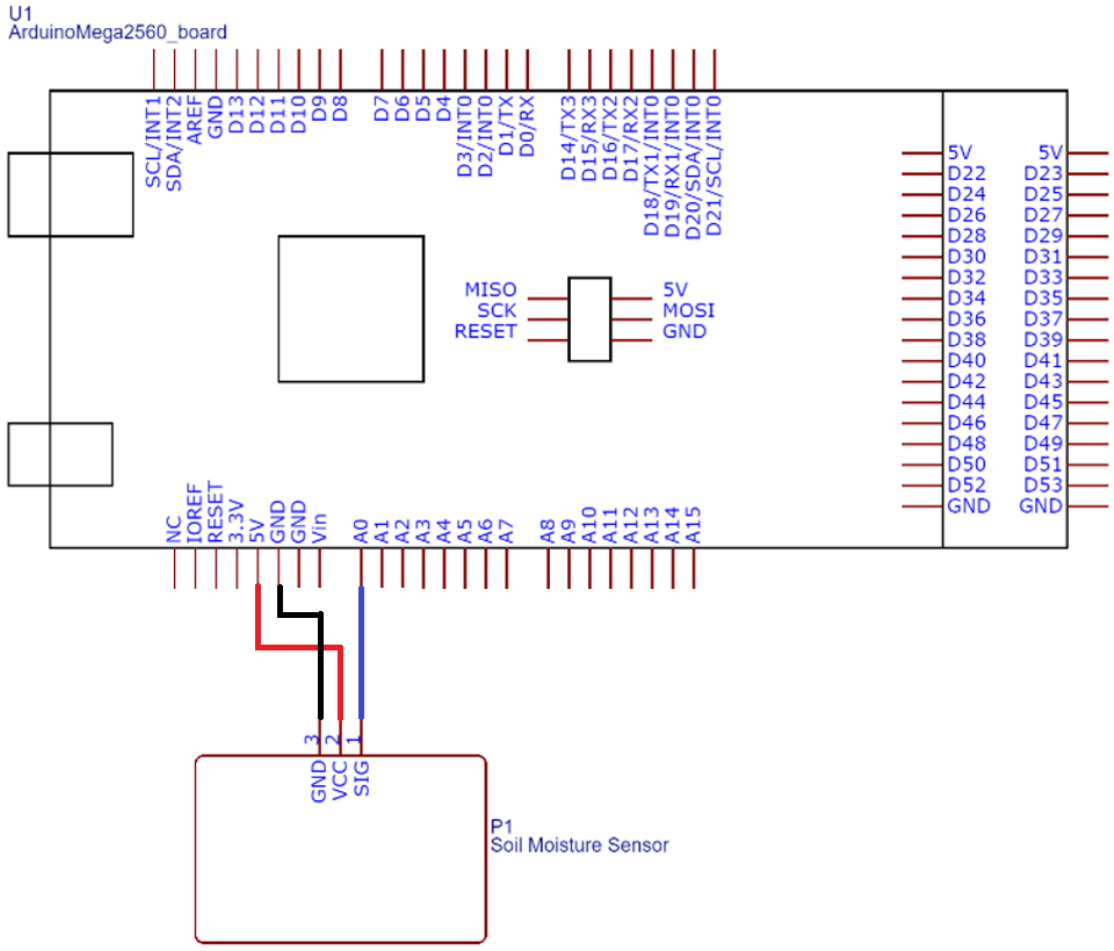


Figure 1: Circuit Diagram

3. Prepare a small cup and fill **1/3 of the cup** with water. This will allow you to test the Capacitive Soil Moisture Sensor. **Make sure that there are no water drops at the top 2/3 of the cup.**

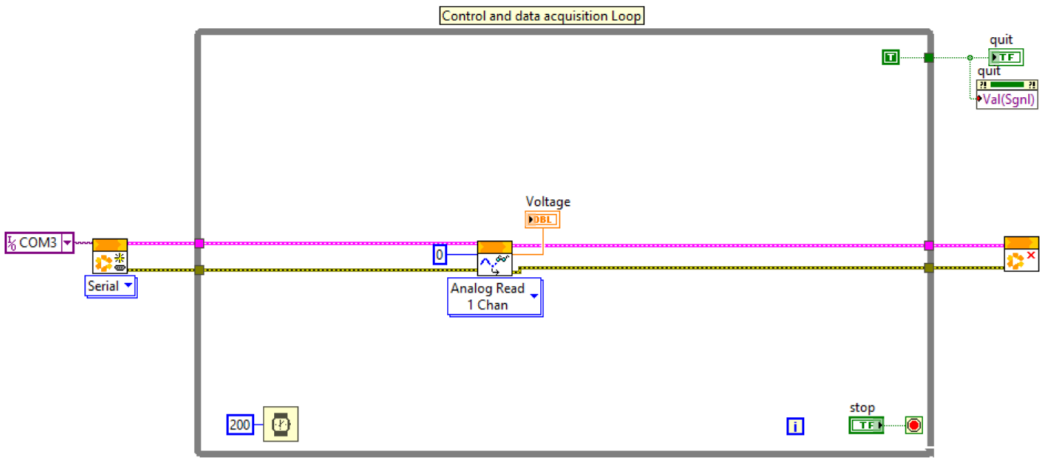
WARNINGS

1. **WHEN WORKING WITH ELECTRONICS, YOUR HANDS AND BODY MUST BE DRY ALL THE TIME. ELECTRIC SHOCKS ARE MUCH MORE SEVERE IF YOUR HANDS ARE WET.**
2. **BE CAREFUL NOT TO SPILL WATER FROM THE CUP ON ANY OF THE ELECTRONIC COMPONENTS OR YOUR LAPTOP. THIS CAN DAMAGE ALL ELECTRONICS.**

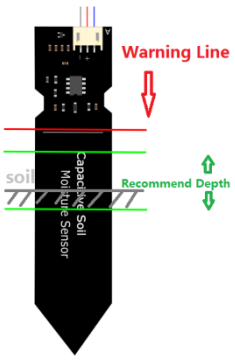
End of Exercise

Exercise 4: LabVIEW program creation

1. Flash the LINX firmware to your Arduino by selecting **Tools»Maker Hub»LINX»LINX firmware Wizard**.
2. Navigate to the `Control and data acquisition Loop`
    a. Add the **LINX Open**, **LINX Close**, and **LINX Analog Read** functions
        i. Make sure to configure the serial port of the **LINX Open** function.
        ii. Make sure to set the analog channel to `0`.
3. Add an indicator to display the voltage of the sensor. Place the indicator in the HMI tab of the front panel. The block diagram of the Control and data acquisition loop should be as follows.

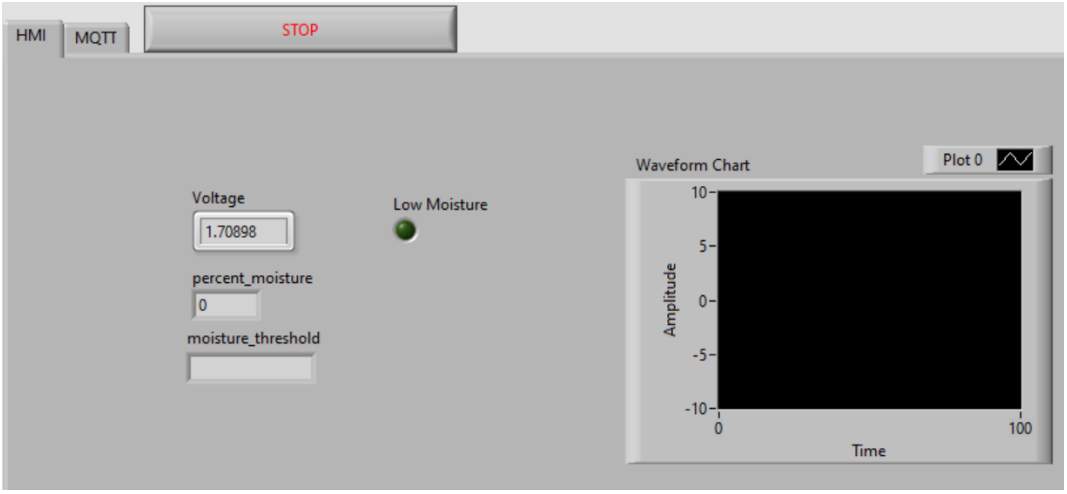


4. **Run** the LabVIEW application **WITHOUT** pressing the **connect** button in the MQTT tab. Monitor the `Voltage` indicator.
    a. When the sensor is in the **air, record** the average voltage reading. This will correspond to 0% moisture (`V_dry`).
    b. Slowly dip the sensor into a cup of **water**, until it touches the bottom of the cup. Record the reading. This corresponds to 100% moisture (`V_wet`).
    WARNING: Do not submerge the circuitry at the top of the sensor into the water. It is not waterproof.



5. Stop your VI and open the block diagram to convert the Voltage into Percent_moisture
6. In the front panel:
    a. Add a string indicator in the HMI tab of the front panel. Name the indicator **moisture_threshold.**
    b. Add a **Round LED** named `Low Moisture` and place it inside the `HMI` tab.
    c. Add a numeric Indicator and a waveform chart for the `percent_moisture`. Place them inside the `HMI` tab of the front panel.

You front panel should look as follows.



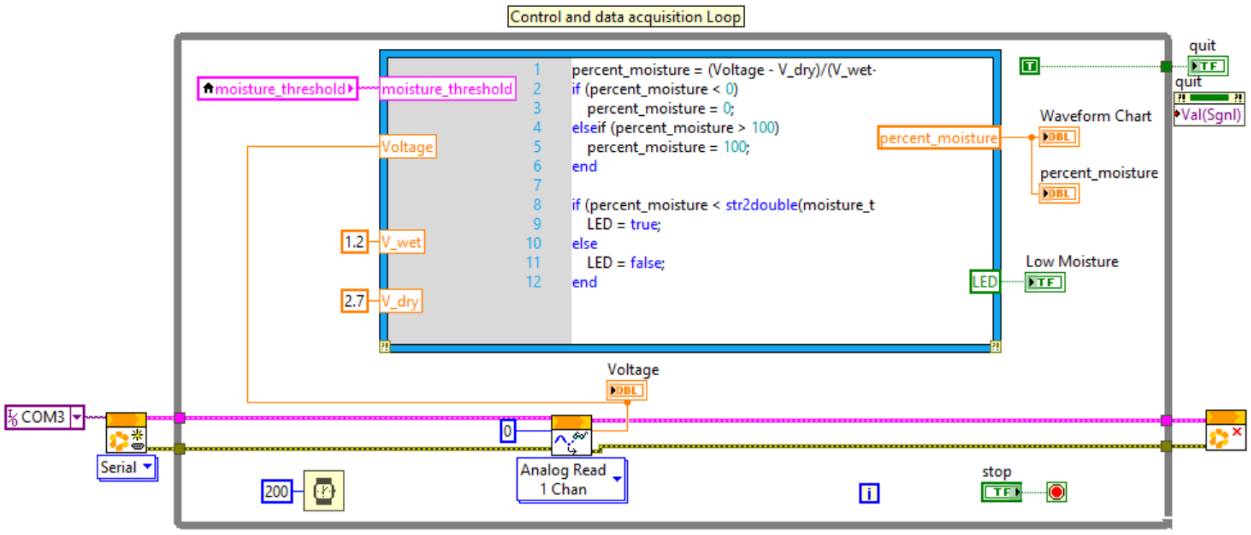
7. In the block diagram:
    a. Add a **Mathscript Block**. This block will convert the analog reading to a percent moisture value from 0 to 100 and activate an alarm if the moisture is below a threshold.
        i. Add 4 inputs: `moisture_threshold`, `Voltage`, `V_wet` and `V_dry`
        ii. Add a **Local Variable** at the left of the Mathscript block. Set the local variable to `moisture_threshold`. Right click and select **change to read**. Connect the local variable to the `moisture_threshold` input of the Mathscript block.
        iii. Connect the `Voltage` input to the voltage output of the Analog Read block.
        iv. Right click `V_wet` and `V_dry` inputs and select create constant. Set the constant values to the values found in step 4.
        v. Paste the following code into the block:

```
percent_moisture = (Voltage - V_dry)/(V_wet-V_dry)*100
if (percent_moisture < 0)
    percent_moisture = 0;
elseif (percent_moisture > 100)
    percent_moisture = 100;
end

if (percent_moisture < str2double(moisture_threshold))
    LED = true;
else
    LED = false;
end
```

        vi. Add 2 outputs to the Mathscript block: `percent_moisture` and `LED`
    b. Connect `percent_moisture` to the waveform chart and percent_moisture indicator.
    c. Connect the `LED` to the Low Moisture round LED.

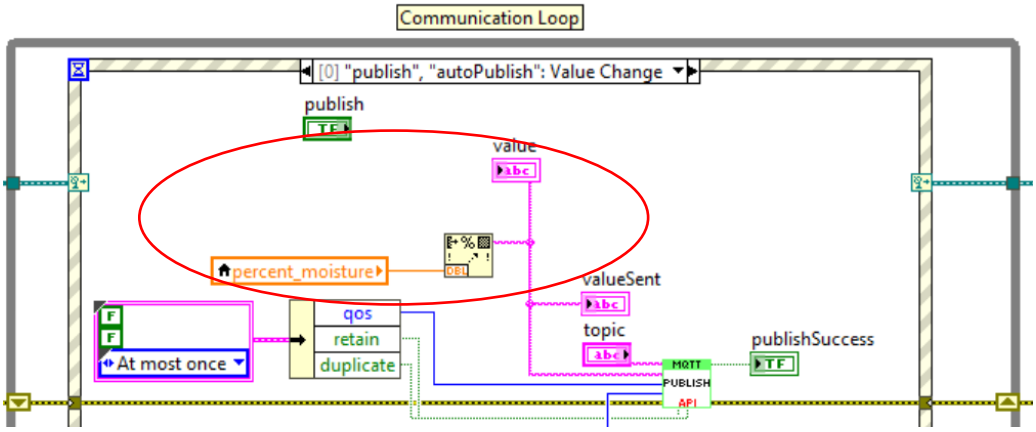
Your block diagram should look as follows.



End of Exercise

Exercise 5: Sending the moisture sensor data to Ubidots IoT platform

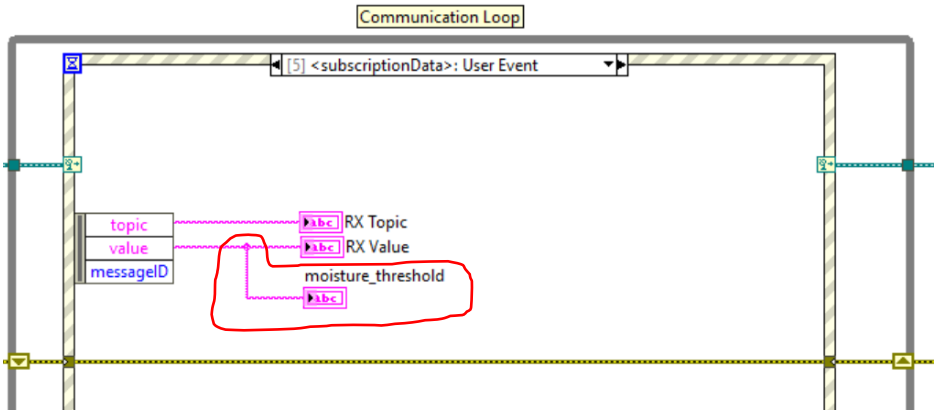
1. In the Communication Loop, go to the `"publish", "autoPublish": Value Change` **Event**.
    a. Add a **Local Variable** `percent_moisture`. Right click and select **change to read**.
    b. Add a **Format into String** function from the **Programming»Strings** palette.
    c. Right click the **value block** and select `change to indicator`.
    d. Connect the blocks as shown below



2. Run LabVIEW. Click on the Connect and Enable Auto Publish push buttons. This will allow to send the sensor data every 1 second.
3. Go to Ubidots, open the moisture variable detailed page (Devices>>Arduino>> moisture), and confirm that the sensor values are continuously streamed to the variable. [Hint: you might need to refresh your page].

Exercise 6: Receiving the moisture threshold from Ubidots IoT platform

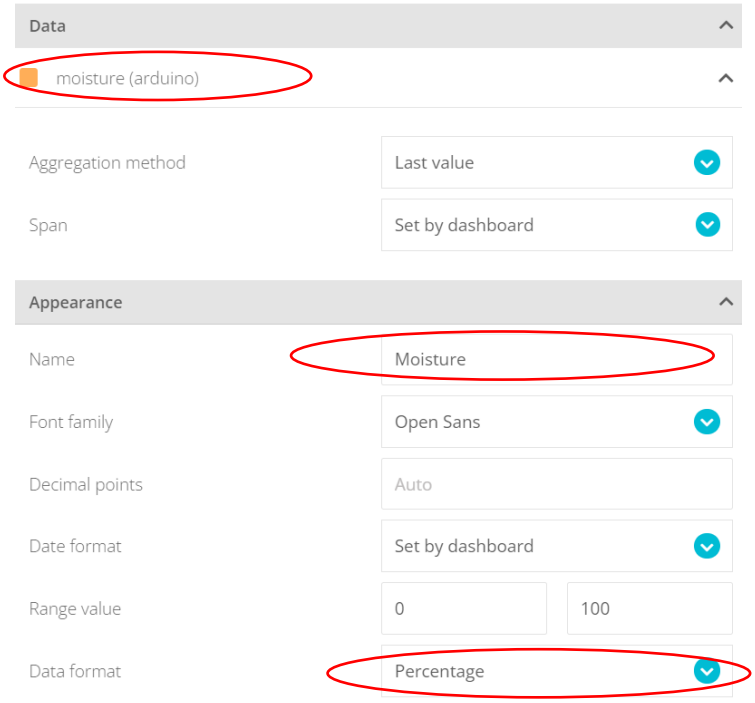
1. In the Communication Loop, go to the `<subscriptionData>: User Event` **Event**.
    a. Wire the moisture_threshold string indicator block as shown below.



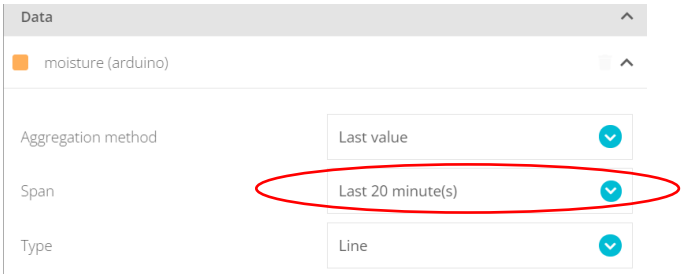
End of Exercise

Exercise 7: Creating a simple dashboard

1. In Ubidots, click on data>>Dashboard.
2. Click on the dashboard menu button  at the top left corner of your screen.
3. Click on the add new dashboard button  to add a new dashboard and name it Lab 8.
4. Click Add new Widget and select a **Tank**. Set the data source to arduino>>moisture; and set the name to moisture.



1. Click Add new Widget and select a **Slider**. Set the data source to arduino>>moisture_threshold; and set the name to Moisture Threshold.
2. Click Add new Widget and select a **Line Chart**. Set the data source to arduino>>moisture; and set the name to Historical Moisture Readings. Set the span to be the Last 20 minute(s) by selecting the custom tab>>Last Values>> Last 20 Minutes.



3. Move and resize your widgets so that your final dashboard will look like:



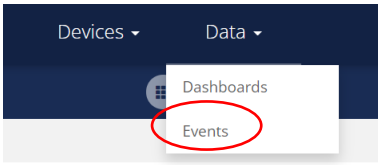
4. Test your setup.
    a. Run the LabVIEW application.
    b. In LabVIEW press **connect** and **enable auto publishing.** The Node-RED application will start receiving MQTT messages.
    c. Set new **moisture thresholds** and click submit. Confirm that the LabVIEW LED will be turned on once the moisture falls below the set moisture threshold.
    d. Move your sensor in and out of the cup and confirm that Ubidots line chart and Tank widgets are updated properly.

End of Exercise

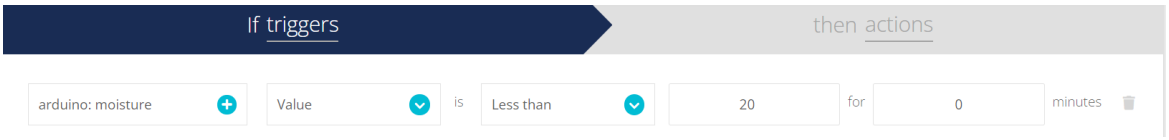
Exercise 7: Adding Triggers to alert users about abnormal conditions

You goal in this exercise is to automatically send an email once the moisture is less than 20%.

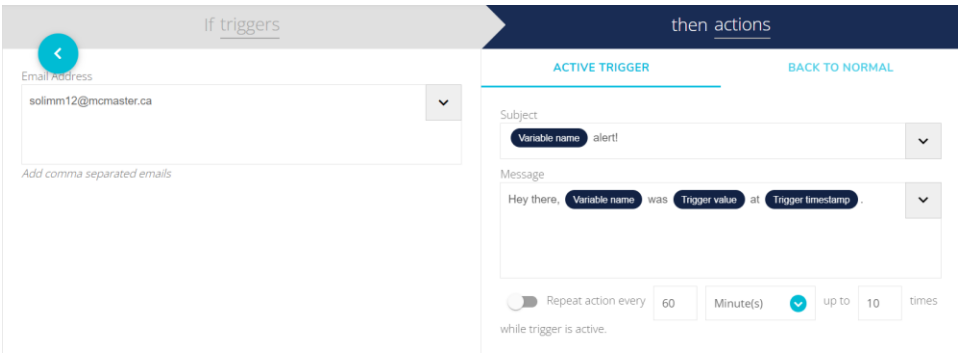
1. Click Data>>Events.



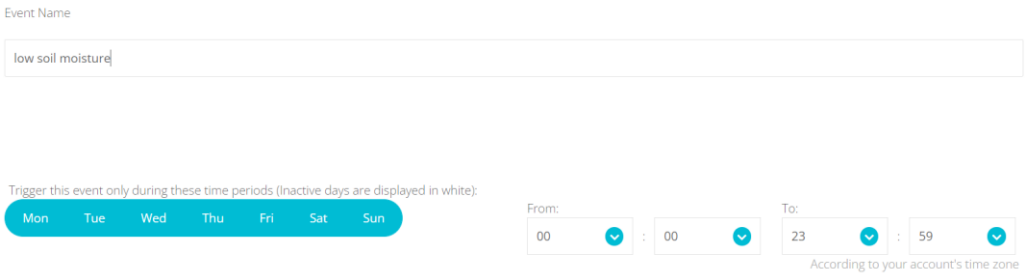
2. Select create event and set the **If triggers** as follows.



3. Click next to configure **then actions**.
4. Add new action>>Send Email.
5. Confirm the email address and customize the email body as you wish.



6. Save the event name to low soil moisture.



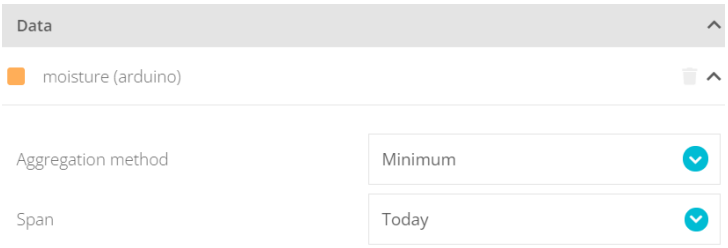
7. Go to dashboard.
8. Test the functionality of the email alert by taking the sensor out of the cup and confirm the receipt of the email.

Note: do not send too many emails since you have 30 free emails in Ubidots STEM account.

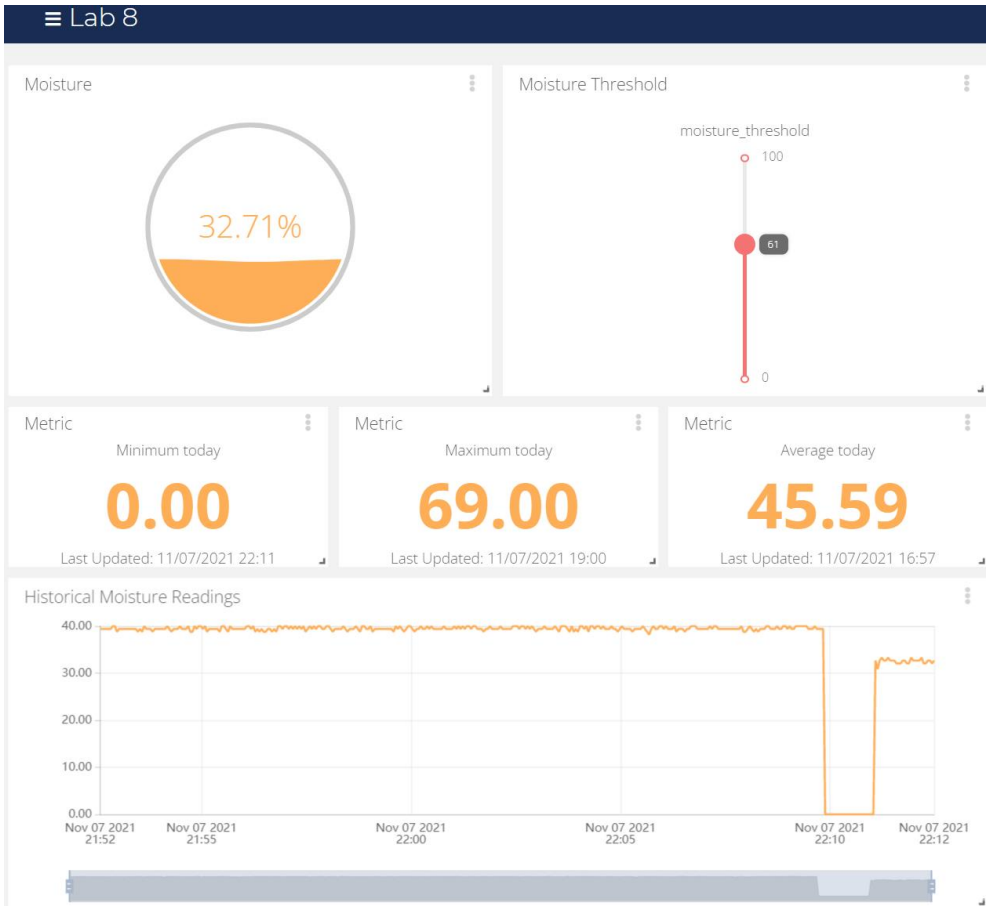
Exercise 8: Adding simple data analytics to the dashboard

You goal in this exercise is to display some daily statistics about the soil moisture.

1. Go to Lab 8 dashboard.
2. Click Add new Widget and select a **Metric**. Set the data source to arduino>>moisture.
3. Set the Aggregation method to Minimum and the span to Today.



5. Repeat step 1 and 2 to add two metric widgets to display the Maximum and the Average Moisture.
6. Rearange your final dashboard to look as follows.



SUBMISSION

1. Draw a block diagram of the system that you built and configured in this lab. In your diagram, use arrows to show how the data are transferred between the web-based HMI and the sensors and actuators.
2. There are two types of soil moisture sensors: capacitive and resistive. Summarize their operation and discuss the advantages of using a capacitive moisture sensor.
3. Explain the Mathscript code in Exercise 4 and why is it important to do step 4 in Exercise 4.
4. Provide one real-life application of using a soil moisture sensor and explain the benefits of making the soil moisture sensor an IoT device for this application.
5. Include any typo that you found or suggestion for improvement for this lab manual.

WANRNING: AFTER COMPLETING THIS LAB,

1. PLEASE TURN OFF THE POWER AND DICONNECT THE ARDUINO FROM THE USB
2. CAREFULLY REMOVE THE SENSOR FROM THE CUP.
3. LET THE SENSOR DRY
4. **DELETE ALL UBIDOTS EVENTS AND PLUGINS (IF USED) AS THE STEM ACCOUNT ALLOWS A VERY LIMITED NUMBER OF THESE ITEMS.**